1 Atmospheric Delay Modelling: Troposphere and Ionosphere

The undifferenced code and phase equations, the code observation and the phase observation of the Foundations document, each carry two propagation terms whose interiors are not states of the filter but *models*: the slant tropospheric delay T_r^i and the first-order slant ionospheric delay $\mu_s\, I_r^i$. The troposphere is split into a deterministic, a-priori *zenith hydrostatic* part that is computed and removed before the filter sees the measurement, plus a small *zenith wet* residual and a pair of horizontal gradients that ARE estimated (the state vector, the ZWD mapping). The ionosphere, in the uncombined parameterisation, is carried as one slant state per satellite (dynamic indexing); the models in this chapter supply its a-priori seed, an optional pseudo-observation that ties the relative slants to a Global Ionosphere Map, and the small second- and third-order corrections that the linear $1/f^2$ scaling cannot represent. This document is the derivation of those interiors: the mapping functions, the pressure reductions, the thin-shell geometry, the geomagnetic field, and the sign conventions that decide whether each term is added to or subtracted from the modelled range.

1.1 The slant tropospheric delay

The neutral atmosphere is non-dispersive at GNSS frequencies, so its delay is identical on code and phase and is absent from any ionosphere-free or geometry-free combination. The standard decomposition splits the slant delay into a hydrostatic ("dry") and a wet zenith delay, each carried to the line of sight by its own elevation mapping function, plus an azimuthal gradient [7]:

$$\boxed{T_r^i = m_{\mathrm{h}}(\varepsilon)\,\mathrm{ZHD} + m_{\mathrm{w}}(\varepsilon)\,\mathrm{ZWD} + m_{\mathrm{g}}(\varepsilon)\big(G_{\mathrm{N}}\cos\alpha + G_{\mathrm{E}}\sin\alpha\big).} \tag{1}$$

Of these, ZHD is supplied a priori and removed from the prefit residual; ZWD, G_{N} and G_{E} are estimated states whose Jacobian columns are exactly the mapping coefficients $m_{\mathrm{w}}(\varepsilon)$ and $m_{\mathrm{g}}(\varepsilon)\{\cos\alpha, \sin\alpha\}$ (`src/core/measurement/MeasurementModel.cpp`; the ZWD mapping). The remainder of this subsection derives each factor.

1.1.1 Saastamoinen zenith hydrostatic delay

The zenith hydrostatic delay is fixed to better than a millimetre by the total surface pressure alone, through the Saastamoinen reduction [6] as coded in `src/core/atmosphere/Troposphere.cpp`:

$$\boxed{\mathrm{ZHD} = \frac{0.0022768\ P}{1 - 0.00266\cos 2\phi - 0.00028\,h_{\mathrm{km}}},} \tag{2}$$

with P the total pressure in millibar at the antenna, ϕ the geodetic latitude, and $h_{\mathrm{km}} = h/1000$ the ellipsoidal height in kilometres (`Troposphere.cpp` line 336, denominator `f_lat` line 328). The numerical coefficient $0.0022768\ \mathrm{m\,mbar^{-1}}$ folds the refractivity constant $k_1 = 77.604\ \mathrm{K\,mbar^{-1}}$ with the dry-gas constant and standard gravity; the denominator is the gravity-variation factor $f(\phi, h)$ that converts the column-integrated delay to its value at the local effective gravity.

When neither a surface met sensor nor the GPT3 grid (below) is available, P is taken from a standard-atmosphere barometric reduction of a configured sea-level pressure P_0 and temperature T_0 (`Troposphere.cpp` lines 41–57):

$$P(h) = P_0\left(1 - \frac{L\,h}{T_0}\right)^{g/(RL)}, \qquad L = 0.0065\ \mathrm{K\,m^{-1}}, \quad \frac{g}{RL} \approx 5.25588, \tag{3}$$

with $g = 9.80665\ \mathrm{m\,s^{-2}}$ and $R = 287.05\ \mathrm{J\,kg^{-1}\,K^{-1}}$ the specific gas constant of dry air. The base $(1 - Lh/T_0)$ is clamped to $[0.1, 2.0]$ and the result to $[100, 1100]$ mbar as a numerical guard.

1.1.2 The Marini mapping form

Both the hydrostatic and wet mappings use the normalised three-parameter continued fraction of Marini, normalised to unity at zenith (`Troposphere.cpp` lines 94–99):

$$m(\varepsilon;\, a, b, c) = \frac{1 + \dfrac{a}{1 + \dfrac{b}{1 + c}}}{\sin\varepsilon + \dfrac{a}{\sin\varepsilon + \dfrac{b}{\sin\varepsilon + c}}}. \tag{4}$$

The numerator is the denominator evaluated at $\varepsilon = 90^\circ$ ($\sin\varepsilon = 1$), so $m(90^\circ) = 1$ exactly; the only thing that distinguishes the hydrostatic from the wet mapping, and one model from another, is the triple (a, b, c). Two families are implemented.

Niell / GMF coefficients. The default mapping interpolates the Niell tables [5] of (a, b, c) given at five reference latitudes $15^\circ, 30^\circ, 45^\circ, 60^\circ, 75^\circ$, each as a mean plus an annually varying amplitude (`Troposphere.cpp` lines 224–245). The seasonal value is

$$v(\phi, t) = \bar{v}(\phi) - \delta v(\phi)\,\cos\frac{2\pi\,(\mathrm{doy} - \mathrm{doy}_0)}{365.25}, \qquad \mathrm{doy}_0 = \begin{cases} 28 & \phi \geq 0 \\ 28 + 182.625 & \phi < 0 \end{cases} \tag{5}$$

(`Troposphere.cpp` lines 207, 269–271). The *minus* sign and the southern-hemisphere half-year phase shift are both load-bearing: the amplitude term is subtracted, and the Southern Hemisphere is put in anti-phase by adding half a year to the phase reference rather than by flipping the sign. The latitude argument is the absolute latitude, linearly interpolated and clamped to $[15^\circ, 75^\circ]$; the wet coefficients carry mean values only (no annual term in this model).

A height correction lifts the hydrostatic mapping from sea level to the station, identical in the Niell and VMF3 paths (`Troposphere.cpp` lines 185–189, 285–289):

$$m_\mathrm{h}(\varepsilon, h) = m_\mathrm{h}(\varepsilon, 0) + \left[\frac{1}{\sin\varepsilon} - m(\varepsilon;\, a_\mathrm{ht}, b_\mathrm{ht}, c_\mathrm{ht})\right] h_\mathrm{km}, \qquad \begin{aligned} a_\mathrm{ht} &= 2.53 \times 10^{-5}, \\ b_\mathrm{ht} &= 5.49 \times 10^{-3}, \\ c_\mathrm{ht} &= 1.14 \times 10^{-3}, \end{aligned} \tag{6}$$

with the same Marini form (4) evaluated at the height triple. Only the hydrostatic mapping receives this correction.

VMF3 coefficients. When configured for VMF3 [4], the wet/hydrostatic a-coefficients come from the GPT3 grid (next paragraph) and the b, c coefficients are synthesised on the fly by a degree- and order-12 spherical-harmonic expansion of their seasonal columns (`src/core/atmosphere/Vmf3.cpp`). A Cartesian-form associated Legendre recursion builds the real and imaginary solid harmonics V_{nm}, W_{nm} at the station co-latitude and longitude (`Vmf3.cpp` lines 46–72), the $91 = \sum_{n=0}^{12}(n+1)$ harmonic terms are accumulated against stored coefficient sets (`Vmf3.cpp` lines 80–92), and each

of $b_{\mathrm{h}}, c_{\mathrm{h}}, b_{\mathrm{w}}, c_{\mathrm{w}}$ is then evaluated as a five-coefficient mean-plus-annual-plus-semiannual harmonic (`Vmf3.cpp` lines 29–32, 97–106):

$$v(\mathrm{doy}) = A_0 + A_1\cos\omega + B_1\sin\omega + A_2\cos 2\omega + B_2\sin 2\omega, \qquad \omega = \frac{2\pi\,\mathrm{doy}}{365.25}. \tag{7}$$

The synthesised (a, b, c) feed the very same Marini fraction (4), followed by the height correction (6). A failed grid query silently falls back to the Niell path so a single bad sample never aborts the run.

1.1.3 GPT3 empirical pressure, temperature and VMF3 a-coefficients

The GPT3 5° grid [4] supplies, per cell, the five harmonic coefficients of pressure, temperature, temperature lapse rate, geoid undulation, and the VMF3 hydrostatic and wet a-coefficients (`src/core/atmosphere/Gpt3.cpp` lines 88–96). Each quantity is the same annual-plus-semiannual harmonic as (7) evaluated per cell (`Gpt3.cpp` lines 40–47):

$$V(\mathrm{doy}) = a_0 + A_1\cos\omega + B_1\sin\omega + A_2\cos 2\omega + B_2\sin 2\omega. \tag{8}$$

The grid value sits at the cell's orthometric reference height H_s; it is carried to the station orthometric height $h_{\mathrm{ortho}} = h - N$ (undulation N) by a barometric pressure reduction and a linear temperature lapse (`Gpt3.cpp` lines 34–37, 151–157):

$$P_{\mathrm{stn}} = P_{\mathrm{grid}}\big(1 - K_{\mathrm{P}}\,\Delta h\big)^{n_{\mathrm{P}}}, \quad T_{\mathrm{stn}} = T_{\mathrm{grid}} + \frac{\mathrm{d}T}{\mathrm{d}z}\,\Delta h, \qquad \begin{aligned} K_{\mathrm{P}} &= 2.26\times10^{-5}\,\mathrm{m}^{-1},\\ n_{\mathrm{P}} &= 5.225, \end{aligned} \tag{9}$$

with $\Delta h = h_{\mathrm{ortho}} - H_s$ and $\mathrm{d}T/\mathrm{d}z$ the grid lapse in $\mathrm{K\,km^{-1}}$ applied to Δh in kilometres. The exponent 5.225 is the same $g/(RL)$ ratio as (3), rounded for the grid. Four corners are read and combined by bilinear interpolation in latitude and (wrapped) longitude (`Gpt3.cpp` lines 146–182); the resulting P_{stn} drives the Saastamoinen ZHD of (2), and the interpolated $a_{\mathrm{h}}, a_{\mathrm{w}}$ drive the VMF3 mapping.

1.1.4 The gradient mapping and a degenerate fallback

The azimuthal gradient in (1) uses the Chen–Herring mapping [1], coded identically in the Jacobian and the prefit residual (`src/core/measurement/MeasurementModel.cpp` lines 1116–1124, `src/core/measurement/StateInitializer.cpp` lines 207–211):

$$\boxed{m_{\mathrm{g}}(\varepsilon) = \frac{1}{\sin\varepsilon\tan\varepsilon + 0.0032}, \qquad T^{\mathrm{grad}} = m_{\mathrm{g}}(\varepsilon)\big(G_{\mathrm{N}}\cos\alpha + G_{\mathrm{E}}\sin\alpha\big).} \tag{10}$$

The two partial derivatives $\partial\mathrm{T}/\partial G_{\mathrm{N}} = m_{\mathrm{g}}\cos\alpha$ and $\partial\mathrm{T}/\partial G_{\mathrm{E}} = m_{\mathrm{g}}\sin\alpha$ are exactly the Jacobian columns of the north/east gradient states (the state vector). Finally, a related cold-start guard applies to the *elevation* (hydrostatic/wet) mapping, not the gradient mapping above: when the receiver position is not yet usable (norm below 10^3 m, or non-finite) the geodetic latitude/height needed by the Niell/VMF3 coefficients is unavailable, so the hydrostatic and wet mappings degrade to the geometry-only Black–Eisner form $1/(\sin\varepsilon + a/(\tan\varepsilon + b))$ (`Troposphere.cpp` lines 152–153) purely so the engine never divides by an undefined elevation during cold start.

1.2 The slant ionospheric delay

The ionosphere is dispersive: to first order its excess phase advance and group delay are $\mp 40.3\,\mathrm{STEC}/f^2$, equal in magnitude and opposite in sign on phase versus code, where STEC is the slant total electron content in $\mathrm{el\,m^{-2}}$. The uncombined model carries this as a per-satellite slant state I_r^i referenced to GPS L1, and projects it onto every signal by the dispersion ratio.

1.2.1 The $1/f^2$ scaling and its sign

For a signal of frequency f on a state referenced to $f_{\text{ref}} = f_{\text{L1}}^{\text{GPS}}$, the dispersion coefficient is

$$\boxed{\mu_s = \frac{f_{\text{ref}}^2}{f_s^2}, \qquad \frac{\partial P_s}{\partial I} = +\mu_s, \qquad \frac{\partial \Phi_s}{\partial I} = -\mu_s,} \tag{11}$$

i.e. the slant-iono Jacobian column is $+\mu_s$ on code and $-\mu_s$ on phase (`src/core/measurement/MeasurementModel.cpp` lines 1145–1149). This is the model interior of μ_s in the notation document; referencing every constellation's slant state to GPS L1 is what lets the between-satellite/between-receiver differencing operators (the differencing document) cancel the receiver iono datum cleanly. Because $f_{\text{ref}} = f_{\text{L1}}$, the L1 signal has $\mu = 1$ and the state *is* the L1 slant delay in metres.

1.2.2 IONEX / GIM thin-shell slant delay

A Global Ionosphere Map gives vertical TEC on a latitude/longitude grid at a single thin shell of height H above a base radius R_E. The slant delay along a ray at elevation ε follows from single-layer geometry (`src/core/atmosphere/GlobalIonosphereMap.cpp` lines 114–161). The zenith angle z' at the ionospheric pierce point (IPP) obeys

$$\sin z' = \frac{R_E}{R_E + H}\cos\varepsilon, \qquad \text{mapping } M(\varepsilon) = \frac{1}{\cos z'}, \tag{12}$$

so that $\text{STEC} = M(\varepsilon)\,\text{VTEC}$ (`GlobalIonosphereMap.cpp` lines 129–132). The earth-centred angle to the IPP is $\psi = \frac{\pi}{2} - \varepsilon - z'$, and the IPP geodetic coordinates come from the spherical-triangle solution along the satellite azimuth α (`GlobalIonosphereMap.cpp` lines 136–140):

$$\phi_{\text{IPP}} = \arcsin\!\big(\sin\phi_r\cos\psi + \cos\phi_r\sin\psi\cos\alpha\big), \tag{13}$$

$$\lambda_{\text{IPP}} = \lambda_r + \arcsin\frac{\sin\psi\,\sin\alpha}{\cos\phi_{\text{IPP}}}. \tag{14}$$

The VTEC at the IPP is interpolated bilinearly in space and, in time, *sun-fixed*: each bracketing map is sampled at the IPP longitude rotated by the Earth's rotation between the map epoch and the query epoch, $\Delta\lambda = (t - t_{\text{map}}) \cdot 360^\circ/86400$ s, before the two are linearly blended (`GlobalIonosphereMap.cpp` lines 16, 89–97). The slant delay in metres at frequency f then uses the dispersion constant

$$\boxed{I^{\text{GIM}} = \text{STEC} \cdot \frac{40.3 \times 10^{16}}{f^2} \quad [\text{m}], \qquad \text{STEC} = M(\varepsilon)\,\text{VTEC [TECU]},} \tag{15}$$

the factor 10^{16} converting VTEC from TECU to $\text{el}\,\text{m}^{-2}$ and $40.3 = 40.3\ \text{m}\,\text{Hz}^2\,(\text{el}\,\text{m}^{-2})^{-1}$ the first-order refractivity constant (`GlobalIonosphereMap.cpp` lines 13–15, 149–153).

1.2.3 The GIM iono-weighting pseudo-observation

The GIM is too coarse to constrain the *absolute* slant of any one satellite, but it constrains the *relative* slants across a constellation well. This is imposed as a Kalman pseudo-observation (`src/core/app/Application.cpp` lines 683–777, `applyGimIonoUpdate`), one row per satellite that owns a slant state above the elevation cutoff. The innovation is the GIM slant minus the current state,

$$z_i = I_i^{\text{GIM}} - I_r^i, \qquad \boldsymbol{H} = \boldsymbol{I} \text{ (selection of the iono columns)}, \tag{16}$$

with the GIM evaluated at GPS L1 to match the state reference (`Application.cpp` lines 728–733). The structure is in the measurement-noise matrix. Per constellation g, with $\mathbf{1}$ the all-ones vector over that constellation's rows,

$$\boxed{\boldsymbol{R} = \bigoplus_{g\in\mathcal{G}} \Big(\operatorname{diag}(\sigma_i^2) + \sigma_c^2\, \mathbf{1}\mathbf{1}^{\mathrm{T}} \Big),} \tag{17}$$

(`Application.cpp` lines 749–760). The rank-one $\sigma_c^2\,\mathbf{1}\mathbf{1}^{\mathrm{T}}$ block assigns an essentially unbounded variance to the per-constellation *common mode* of the innovations: that common mode is the un-modellable receiver iono datum (an arbitrary offset shared by all satellites of a constellation), so the pseudo-observation leaves it free and constrains only the between-satellite differences that the GIM actually knows. The update is applied through the same Joseph-form pseudo-update as every other measurement (the Kalman filter document); the iono process noise is documented there (the iono random-walk spectral density) and not repeated here.

Why frozen pivots are excluded. In the BSSD parameterisation each constellation's pivot satellite holds its slant-iono state frozen at value 0 with variance 0 as the rank-fixing datum. Such a row is skipped entirely (`Application.cpp` lines 707–725). A zero-variance state receives zero Kalman gain and so could never benefit from the GIM; but it shares the common-mode $\mathbf{1}\mathbf{1}^{\mathrm{T}}$ block of (17) with its constellation peers. Were it included, its spurious innovation $I_{p_g}^{\mathrm{GIM}} - 0 \sim 5$–$30$ m would couple, through the off-diagonal of $\boldsymbol{S}^{-1}$, into the gain of every other satellite of that constellation, dragging the relative slants by the pivot's full absolute STEC. Excluding the pivot row is therefore not an optimisation but a correctness requirement: the common-mode block is meant to absorb the datum offset of the *remaining* rows, so the datum satellite must not itself appear as a measurement.

1.2.4 Klobuchar broadcast seed

For a single-frequency a-priori only — the cold-start seed of the slant state, never a dual-frequency correction — the broadcast Klobuchar half-cosine model is available [3] (`src/core/atmosphere/Ionosphere.cpp` lines 151–195). With elevation and azimuth in semicircles, the earth-centred angle, geomagnetic IPP latitude, local time and obliquity are

$$\psi = \frac{0.0137}{\varepsilon/180 + 0.11} - 0.022, \qquad \phi_i = \operatorname{clip}_{\pm 0.416}\big(\phi/180 + \psi\cos\alpha\big), \tag{18}$$

$$\phi_m = \phi_i + 0.064\cos\big((\lambda_i - 1.617)\pi\big), \qquad F = 1 + 16\,(0.53 - \varepsilon/180)^3, \tag{19}$$

with $\lambda_i = \lambda/180 + \psi\sin\alpha/\cos(\phi_i\pi)$ and local time $t_i = 43200\,\lambda_i + t \pmod{86400}$. Cubic polynomials in the geomagnetic latitude ϕ_m give the amplitude and period,

$$\mathrm{AMP} = \sum_{k=0}^{3} a_k\,\phi_m^k \ (\geq 0), \qquad \mathrm{PER} = \sum_{k=0}^{3} b_k\,\phi_m^k \ (\geq 72000\ \mathrm{s}), \tag{20}$$

and the zenith delay is the truncated half-cosine, mapped to slant by F and scaled to the requested frequency by the f_{L1}^2/f^2 dispersion ratio of (11) (`Ionosphere.cpp` lines 183–194):

$$I^{\mathrm{Klob}} = c\,F\Big(5\times10^{-9} + \mathrm{AMP}\,\big[1 - \tfrac{X^2}{2} + \tfrac{X^4}{24}\big]\Big)\frac{f_{\mathrm{L1}}^2}{f^2}, \quad X = \frac{2\pi(t_i - 50400)}{\mathrm{PER}}, \ |X| < 1.57, \tag{21}$$

and the constant 5×10^{-9} s night term alone when $|X| \geq 1.57$. The leading c converts the modelled signal delay from seconds to metres.

1.3 Higher-order ionospheric corrections

The $1/f^2$ scaling captures the first-order term; the residual second- and third-order delays scale as $1/f^3$ and $1/f^4$ and are applied as deterministic corrections to the modelled range (`src/core/atmosphere/HigherOrderIonosphere.cpp` lines 45–103). The second order is the magnetic-field–weighted delay

$$\boxed{I_2^P = 7527\, c\,\mathrm{STEC}\,\frac{\boldsymbol{B}\cdot\hat{\boldsymbol{k}}}{f^3}, \qquad I_2^\Phi = -\tfrac{1}{2}\, I_2^P,} \tag{22}$$

where STEC is in $\mathrm{el\,m^{-2}}$, $\boldsymbol{B}$ is the geomagnetic flux density in tesla at the IPP, $\hat{\boldsymbol{k}}$ the receiver→satellite unit vector, and c the speed of light (the constant $7527\, c = e\,A/(2\pi m_e) \approx 2.2564 \times 10^{12}$ folds the electron charge, the 40.3-family refractivity constant A, and the electron mass; `HigherOrderIonosphere.cpp` lines 10–24, 88). The phase carries exactly $-\frac{1}{2}$ of the code value (`HigherOrderIonosphere.cpp` line 98). The third order depends on the path integral of N_e^2, which a shape factor η relates to the F2-peak density and STEC (`HigherOrderIonosphere.cpp` lines 29–42, 87–89):

$$\boxed{I_3^P = 2437\,\eta\, N_{\max}\,\frac{\mathrm{STEC}}{f^4}, \qquad N_{\max} = \kappa\,\mathrm{STEC}, \qquad I_3^\Phi = -\tfrac{1}{3}\, I_3^P,} \tag{23}$$

with $\eta = 0.66$ the Hoque–Jakowski shape factor [2], $2437 = 3A^2/8$, and $N_{\max} = \kappa\,\mathrm{STEC}$ the linear Hoque–Jakowski fit ($\kappa = (20-6)\times 10^{12}/((4.55-1.38)\times 10^{18})$ $\mathrm{el\,m^{-3}/(el\,m^{-2})}$) of the F2-peak electron density to slant STEC. The third-order term is thus *linear* in $N_{\max}\,\mathrm{STEC}$, not in STEC^2. Both corrections are sub-millimetre at L1 but enter with the correct sign on each measurement type and survive between-satellite differencing because they are satellite-specific.

1.3.1 The geomagnetic dipole field

The field at the IPP is a centred, tilted magnetic dipole evaluated in ECEF (`src/core/atmosphere/MagneticField.cpp` lines 33–49):

$$\boxed{\boldsymbol{B}(\boldsymbol{r}) = B_0\left(\frac{R_E}{r}\right)^3\left(\hat{\boldsymbol{m}} - 3(\hat{\boldsymbol{m}}\cdot\hat{\boldsymbol{r}})\,\hat{\boldsymbol{r}}\right),} \tag{24}$$

with $r = \|\boldsymbol{r}\|$, $\hat{\boldsymbol{r}} = \boldsymbol{r}/r$, $B_0 = 30 \times 10^{-6}$ T the equatorial surface field, $R_E = 6371.2$ km the IGRF reference radius, and $\hat{\boldsymbol{m}}$ the unit axis pointing to the geographic location of the geomagnetic north pole, IGRF-13 epoch 2020.0 at $(80.65°, -72.68°)$ (`MagneticField.cpp` lines 14–31). With $\hat{\boldsymbol{m}}$ directed at the Arctic geomagnetic pole (which is the dipole's *south* pole), the sign convention in (24) makes $\boldsymbol{B}$ point horizontally toward magnetic north at the magnetic equator, matching compass behaviour; this sign is what fixes the sign of $\boldsymbol{B}\cdot\hat{\boldsymbol{k}}$ and hence of I_2 in (22). The IPP ECEF position itself is reconstructed as a spherical-to-Cartesian map of the geocentric IPP latitude/longitude at the IONEX base-plus-shell radius, since IONEX is a thin-shell model (`HigherOrderIonosphere.cpp` lines 61–69).

Together with the deterministic ZHD removal, the estimated ZWD/gradient states, and the uncombined slant-iono state, these models populate every atmospheric term of the undifferenced observation equation; the differencing operators that subsequently eliminate the receiver-common pieces of them are the subject of the differencing document.

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